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Ex 1: Salaries of entry-level computer engineers

The salaries of entry-level computer engineers are assumed to follow a normal distribution with unknown mean and unknown variance. A random sample of 12 engineers gave the following monthly salaries (in $):

1074.67	1011.17	1089.77	1177.30	1001.58	1001.59
1182.92	1101.74	978.05	1079.26	978.66	978.43

Sample statistics used in all computations:

Statistic	**Symbol**	**Value**
Sample size	n	12
Sample mean	$\bar{x}$	1054.60 $
Sample standard deviation (n − 1 in the denominator)	s	74.42 $
Standard error of the mean (s / √n)	SE	21.48 $

All intervals use a confidence level of 90%, so $\alpha = 0.10$ and each tail contains $\alpha/2 = 0.05$. The critical values of the Student distribution were read from Tabelul B (quantiles of order $1 - \alpha/2$); the chi-square quantiles were computed with SciPy.

a) 90% confidence interval for the average salary

Since the population variance is unknown and the sample is small, the interval for the mean is based on the Student t distribution with $n - 1 = 11$ degrees of freedom:

$$\bar{x} \pm t_{1-\alpha/2,\, n-1} \cdot s / \sqrt{n}$$

The critical value is read from Tabelul B at row n = 11 and column $1 - \alpha/2 = 0.950$, which gives t = 1.796. The margin of error is the critical value times the standard error of the mean:

```
Table row (df) = 11, column = 0.950  ->  t critical value = 1.796
Standard error = 21.4842, margin of error = 38.5856
90% CI for the mean: [1016.01, 1093.18]
SciPy check: (np.float64(1016.0118547509272), np.float64(1093.178145249073))
```

Interpretation

We are 90% confident that the average salary of an entry-level computer engineer lies between about 1016 $ and 1093 $ per month. The confidence level refers to the method: if samples of 12 engineers were drawn repeatedly, about 90% of the intervals built in this way would contain the true mean. The margin of error (38.59 $) is the distance from the sample mean to each end of the interval; it grows with the variability of the data (s) and with the confidence level, and it shrinks as the sample size increases. The same interval computed with the exact t quantile in SciPy is identical to two decimals, the only difference being the rounding of the table.

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b) 90% confidence interval for the standard deviation

For a normal population, the quantity $(n-1)s^2 / \sigma^2$ follows a chi-square distribution with $n - 1 = 11$ degrees of freedom. This gives the interval for σ:

$$\left[\sqrt{(n-1)s^2 / \chi^2_{1-\alpha/2}} \, , \; \sqrt{(n-1)s^2 / \chi^2_{\alpha/2}} \right]$$

Quantity	**Value**
$(n-1)\cdot s^2$	$11 \times 74.42^2 = 60\,927.3$
χ^2 quantile of order 0.05 (df = 11)	4.575
χ^2 quantile of order 0.95 (df = 11)	19.675
Lower limit √(60 927.3 / 19.675)	55.65
Upper limit √(60 927.3 / 4.575)	115.40
90% CI for σ	[55.65 ; 115.40]

```
90% CI for the standard deviation: [55.65, 115.40]
```

Interpretation

The standard deviation of the starting salaries is estimated to lie between about 55.65 $ and 115.40 $. The interval is not symmetric around s = 74.42 $ (it extends 18.8 $ below and 41.0 $ above) because the chi-square distribution is skewed to the right. It is also very wide: the upper limit is more than twice the lower limit. With only 12 observations the variability is estimated imprecisely, so the first impression that the salaries have “a great deal of variability” cannot be confirmed or rejected reliably from this sample alone.

c) New sample of 100 engineers with the same mean and standard deviation

Assuming a new sample of size n = 100 with the same $\bar{x}$ = 1054.60 and s = 74.42, both intervals were recomputed. For the mean, df = 99 is not tabulated, so the nearest row of Tabelul B (n = 100, column 0.950) was used, giving t = 1.660. The standard error becomes 74.42 / √100 = 7.44 and the margin of error 1.660 × 7.44 = 12.35. For the standard deviation, the chi-square quantiles with 99 degrees of freedom were used (77.05 and 123.23).

```
Table row (df) used for n = 100: 100  ->  t critical value = 1.66
90% CI for the mean (n=100): [1042.24, 1066.95]
90% CI for the std  (n=100): [66.71, 84.36]
```

	n	Parameter	Lower	Upper	Width
0	12	Mean	1016.01	1093.18	77.17
1	100	Mean	1042.24	1066.95	24.71
2	12	Std. deviation	55.65	115.40	59.76
3	100	Std. deviation	66.71	84.36	17.65

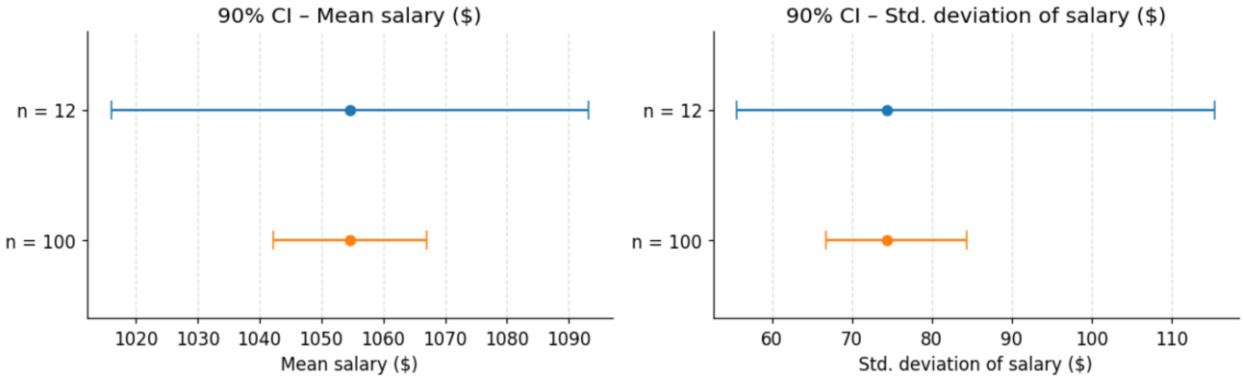


Figure 1. 90% confidence intervals for the mean and the standard deviation, n = 12 versus n = 100

Interpretation

The point estimates are the same, but the intervals for n = 100 are much narrower: the interval for the mean is about 3.1 times shorter (24.71 $ versus 77.17 $) and the interval for the standard deviation about 3.4 times shorter (17.65 $ versus 59.75 $). For the mean, the width decreases roughly with $1/\sqrt{n}$, because the standard error shrinks and the critical value gets closer to the normal one (1.796 versus 1.660). For the standard deviation, the interval also becomes tighter and more symmetric around s, since the chi-square distribution approaches a normal shape as the degrees of freedom grow. In conclusion, a larger sample does not change what we estimate, but it makes the estimates of both the average salary and its variability considerably more precise.

Ex 2: NHANES – height and body weight

The NHANES (2015–2016) dataset is used. It originates from the National Center for Health Statistics (USA) and is part of the National Health and Nutrition Examination Survey (https://wwwn.cdc.gov/nchs/nhanes/); the descriptions of the variables are available at https://wwwn.cdc.gov/nchs/nhanes/search/default.aspx. The variables studied are BMXHT (height, cm), BMXWT (body weight, kg) and RIAGENDR (gender, coded 1 = male, 2 = female).

a) Loading the data, selecting the variables and handling missing values

NHANES data are published as separate SAS transport files, one for each survey component, which share the participant identifier SEQN. The gender variable (RIAGENDR) is stored in the Demographics file (DEMO_I), while height and body weight (BMXHT, BMXWT) are stored in the Body Measures file (BMX_I). Both files for the 2015–2016 cycle were downloaded from the CDC website, read with pandas (read_sas) and merged on SEQN, keeping the participants present in both files.

Only the three variables of interest (RIAGENDR, BMXHT, BMXWT) were then extracted. The number of missing entries (NaN) in each column was checked with isna().sum(), and the rows containing missing values were removed with dropna(), so that all later computations use complete observations only. Finally, the gender code, which is stored as a decimal number in the transport file, was converted to an integer (1 = male, 2 = female).

```
New shape: (9544, 3)

Missing values per column:
RIAGENDR      0
BMXHT       775
BMXWT        99
dtype: int64

Shape after removing NaN: (8756, 3)

Participants per gender:
Gender
Female    4470
Male      4286
Name: count, dtype: int64
```

	RIAGENDR	BMXHT	BMXWT	Gender
0	1	184.5	94.8	Male
1	1	171.4	90.4	Male
2	1	170.1	83.4	Male
3	2	160.9	109.8	Female
4	2	164.9	55.2	Female

b) 95% and 99% confidence intervals for the mean

The population variance is unknown, so the intervals are $\bar{x} \pm t \cdot s/\sqrt{n}$, with the critical value read from Tabelul B. All groups (males, females and the whole sample) contain far more than 100 observations, so the last row of the table (∞) applies: $t = 1.960$ for 95% (column 0.975) and $t = 2.576$ for 99% (column 0.995). The intervals were computed separately for each gender and for the entire sample, for both variables.

	Confidence	Variable	Group	n	Mean	Lower	Upper	Width
0	95%	Height (cm)	Male	4286	159.660	158.909	160.412	1.503
1	95%	Height (cm)	Female	4470	150.680	150.099	151.261	1.162
2	95%	Height (cm)	All	8756	155.076	154.594	155.558	0.963
3	95%	Weight (kg)	Male	4286	69.706	68.762	70.650	1.887
4	95%	Weight (kg)	Female	4470	63.626	62.809	64.443	1.634
5	95%	Weight (kg)	All	8756	66.602	65.976	67.227	1.251
6	99%	Height (cm)	Male	4286	159.660	158.673	160.648	1.975
7	99%	Height (cm)	Female	4470	150.680	149.917	151.444	1.527
8	99%	Height (cm)	All	8756	155.076	154.443	155.709	1.266
9	99%	Weight (kg)	Male	4286	69.706	68.466	70.946	2.480
10	99%	Weight (kg)	Female	4470	63.626	62.552	64.699	2.148
11	99%	Weight (kg)	All	8756	66.602	65.780	67.424	1.644

Variable	Group	95%	99%	Ratio 99%/95%
Height (cm)	All	0.963	1.266	1.314
	Female	1.162	1.527	1.314
	Male	1.503	1.975	1.314
Weight (kg)	All	1.251	1.644	1.314
	Female	1.634	2.148	1.314
	Male	1.887	2.480	1.314

Interpretation

Raising the confidence level from 95% to 99% makes every interval wider: the critical value increases from 1.960 to 2.576, so each interval is about 1.31 times longer, while the centre (the sample mean) does not change. This is the price of a higher confidence: we are more sure that the interval contains the true mean, but the estimate is less precise. The intervals for the entire sample are narrower than those for each gender, because the standard error s/√n decreases as the number of observations grows. Since all groups are large, every interval is very narrow, so the means are estimated very precisely.

c) Twenty random samples of size 100

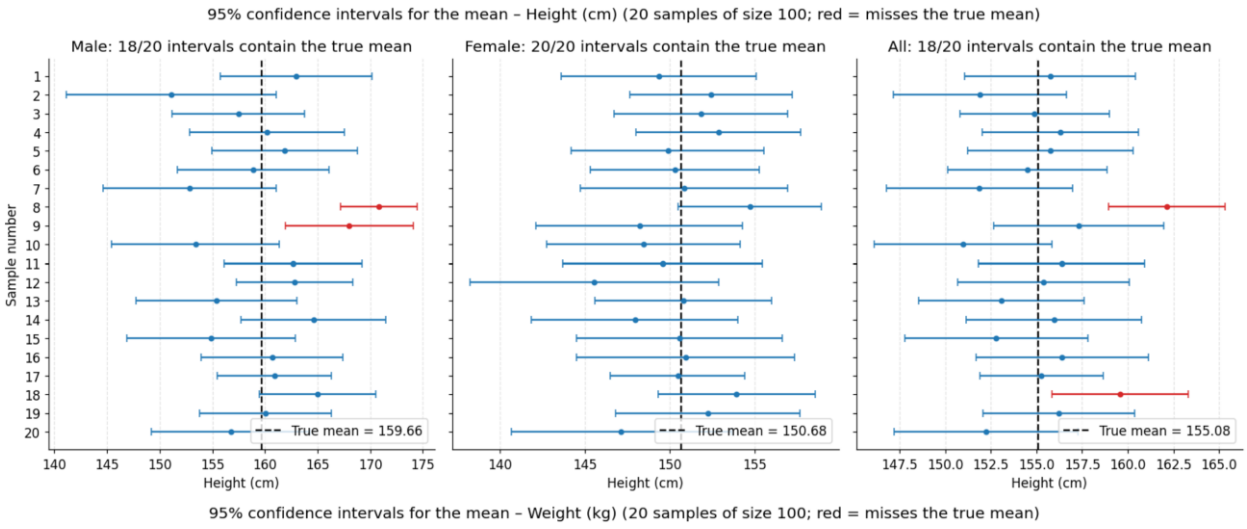
The entire cleaned dataset was treated as the statistical population, so its means (obtained in part b) are the true values. Twenty simple random samples of size 100 were drawn without replacement (fixed seed for reproducibility). For each sample, 95% confidence intervals for the mean height and the mean weight were computed separately for each gender (each gender subsample has about 50 observations) and for the whole sample of 100. The critical values come from Tabelul B: for the whole sample df = 99, so the nearest row (100) gives t = 1.984; for a gender subsample with about 50 observations df ≈ 49, so the nearest row (50) gives t = 2.009.

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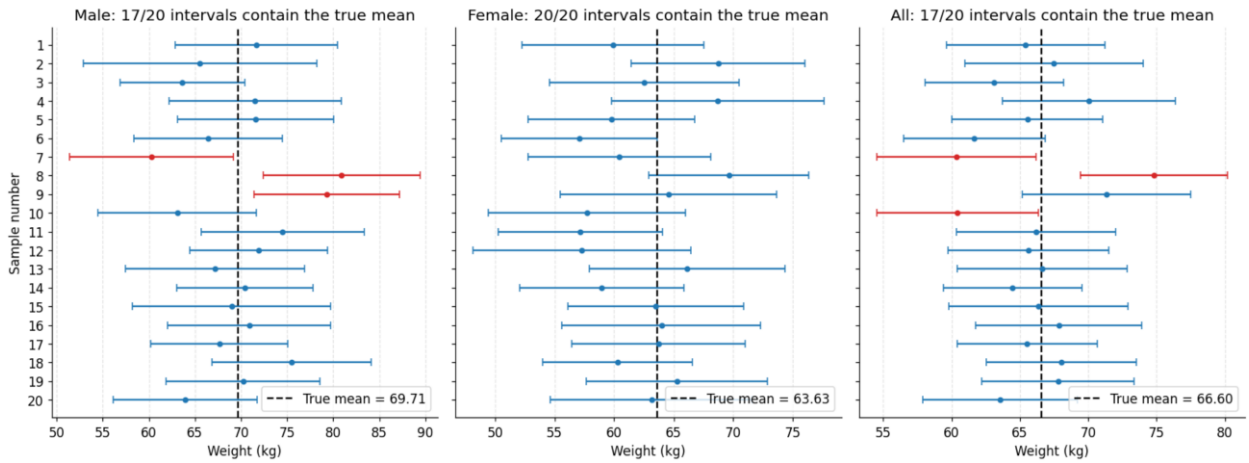
	Sample	Variable	Group	n	Mean	Lower	Upper	Covers true mean
0	1	BMXHT	Male	47	162.962	155.782	170.141	True
1	1	BMXHT	Female	53	149.347	143.604	155.090	True
2	1	BMXHT	All	100	155.746	151.080	160.412	True
3	1	BMXWT	Male	47	71.660	62.833	80.487	True
4	1	BMXWT	Female	53	59.900	52.257	67.543	True
5	1	BMXWT	All	100	65.427	59.611	71.243	True
6	2	BMXHT	Male	39	151.095	141.128	161.062	True
7	2	BMXHT	Female	61	152.416	147.637	157.196	True
8	2	BMXHT	All	100	151.901	147.141	156.661	True

Interpretation

Each sample gives a different sample mean, so the 20 intervals are located at different places around the true mean (dashed line); this is the sampling variability of the estimator. A 95% confidence level means that, in the long run, about 95% of the intervals contain the true mean, so out of 20 intervals we expect about one (19 out of 20) to miss it. Results such as 18–20 out of 20 are consistent with this, and small deviations are normal because only 20 repetitions were made. The intervals for the gender groups are wider than those for the whole sample, because each gender subsample has only about 50 observations. All of these intervals are also much wider than those from part b, which used the entire dataset: the precision of the estimate is controlled by the sample size.



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Variable	Group	Intervals covering the true mean (of 20)	Coverage rate
BMXHT	All	18	0.90
	Female	20	1.00
	Male	18	0.90
BMXWT	All	17	0.85
	Female	20	1.00
	Male	17	0.85

Comments (2c)

- For each sample the critical value comes from Tabelul B: df = 99 → row 100 (1.984) for the whole sample and, for each gender subsample of about 50 observations, the nearest row (e.g. df = 49 → row 50 → 2.009).
- Each of the 20 intervals is centred on a different sample mean, so the intervals move around the true mean (dashed line) from sample to sample: this is sampling variability.
- A 95% confidence level means that, in the long run, about 95% of the intervals contain the true mean, so about 1 interval out of 20 is expected to miss it (red). The coverage table shows how many actually do; a result of 18–20 out of 20 is in line with theory, and small deviations are due to the small number of repetitions (only 20).
- The intervals for the gender groups are wider than those for the entire sample, because each gender subsample has only about 50 observations.
- Intervals computed on samples of size 100 are much wider than those computed from the entire dataset in 2b, which shows how the sample size controls precision.
- Changing the seed in np.random.default_rng(42) gives different samples and possibly different numbers of intervals missing the true mean.